



Compressible Turbulent Boundary Layer with Pressure Gradient and Heat Transfer

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The compressible turbulent boundary layer is expressed in terms of the momentum integral and moment-of-momentum integral equations for arbitrary pressure gradient. Upon simplification of this equation set with a Mager-type transformation, the two coupled differential equations are solved simultaneously to provide boundary-layer momentum thickness, displacement thickness, skin-friction coefficient, and related quantities as a function of the body surface coordinate. The analysis includes surface heat-transfer effects. Results of the analysis are presented for adiabatic, non-adiabatic, two-dimensional, and axisymmetric flow-fields in a Mach number range between 2 and 8. Flows with favorable and adverse pressure gradients are also examined. The present analysis is compared with available experimental and theoretical results. The analysis is found to yield satisfactory results for flows with and without heat transfer. In addition, for flows in the pressure of large adverse pressure gradients in which the boundary layer is known to be well-behaved, spurious separation is not indicated; instead, rather good agreement with available experimental data is observed.

Nomenclature

a = speed of sound
 C_f = local skin-friction coefficient
 C_p = coefficient of specific heat at constant pressure
 g = normalized enthalpy, h/h_0
 h = enthalpy
 H = form factor, δ^*/θ
 H_i = transformed plane form factor for adiabatic flow
 H_{tr} = transformed plane form factor for nonadiabatic flow, Δ^*/θ
 i = 0, 1 for 2-dimensional and axisymmetric flow, respectively
 k = coefficient of thermal conductivity
 L = reference length, 1 in.
 M = Mach number
 N = exponent of power law profile, $(y/\delta)^N$, for the physical plane
 N_t = exponent of power law profile, $(Y/\Delta)^{N_t}$, for the transformed plane
 Nu = Nusselt number, $q_w C_{p0} L / k_0 (h_0 - h_w)$
 p = pressure
 q = local heat-transfer rate
 R = radius of axisymmetric body
 T = temperature
 u = longitudinal velocity
 u_τ = friction velocity, $(\tau_w/\rho)^{1/2}$
 U = transformed longitudinal velocity
 v = normal velocity
 x = coordinate along body surface
 X = transformed coordinate along body surface
 y = normal coordinate
 Y = transformed normal coordinate
 γ = ratio of specific heats
 Δ = transformed boundary-layer thickness
 δ^* = displacement thickness, $\int_0^\delta \left(1 - \frac{\rho u}{\rho_e u_e}\right) dy$
 Δ^* = transformed displacement thickness, $\int_0^\Delta \left[\left(1 - \frac{U}{U_e}\right) + (g - 1) \right] dY$
 θ = momentum thickness, $\int_0^\delta \frac{\rho u}{\rho_e u_e} \left(1 - \frac{u}{u_e}\right) dy$

Θ = transformed momentum thickness, $\int_0^\Delta \frac{U}{U_e} \left(1 - \frac{U}{U_e}\right) dY$
 μ = viscosity coefficient
 ν = kinematic viscosity coefficient, μ/ρ
 ρ = mass density
 σ = Prandtl number, $C_p \mu / k$
 τ = shearing stress
 ψ = stream function
 Ψ = transformed stream function

Subscripts

e = local external flow conditions
 s = local stagnation conditions
 w = wall or surface conditions
 o = freestream stagnation conditions

Superscripts

($\bar{}$) = evaluated at reference conditions, Eq. (9)

I. Introduction

THE turbulent boundary layer has long been the subject of investigation in fluid mechanics. In recent years the understanding of detailed phenomena relating thereto has been advanced significantly with concepts such as the law of the wall and the law of the wake. However, despite these new techniques, a general treatment of the compressible turbulent boundary layer employing these methods has yet to be accomplished.

From a pragmatic point of view, it is highly desirable to couch the solution to the forementioned problem only in terms of known variables of body geometry and the associated inviscid flowfield conditions, rather than to appeal to parameters, such as the velocity defect, which have a fundamental importance in turbulent boundary-layer analysis. Indeed, many investigators, motivated on grounds of practicality and attracted by the apparent conceptual simplicity, have expressed their analyses and results of turbulent boundary-layer behavior in terms of the boundary-layer form factor (H) and momentum thickness (θ). It should be observed, however, that the boundary-layer form factor can be regarded as something less than an ideal parameter for turbulent boundary-layer analysis, since it is generally an unestablished function of pressure distribution and skin friction.¹ Nonetheless, withstanding such apparent shortcomings, in the past a significant amount of work has been accomplished with the aid of the form factor parameter. For incompressible flows, for example, Von Doenhoff and Tetervin² used the

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momentum equation together with an empirical auxiliary equation to calculate boundary-layer form factor distribution. More recently, Tetervin and Lin³ systematically expressed the boundary-layer momentum, moment-of-momentum, and energy integral equations using the form factor concept and applied these results with reasonable success in the prediction of turbulent separation on subsonic airfoils. Maskell⁴ has also extended this process of turbulent boundary-layer analysis by describing the variation of H with x by a series of empirical functions that conveniently uncouple the variation of H from the momentum integral equation.

For turbulent boundary layers in compressible flow, Reshotko and Tucker⁵ again used the momentum and moment-of-momentum integral equations in terms of the form factor and the momentum thickness. The Stewartson transformation was used together with the results of Maskell to express the equations in an incompressible form and to uncouple the two governing equations. This analysis resulted in equations that should tentatively describe turbulent boundary layers in compressible flow with heat transfer. The Reshotko-Tucker analysis, which has enjoyed success for favorable and mild adverse pressure gradient, however, possesses some shortcomings when applied to flows with large adverse gradients. One of the more serious of these is an indicated boundary-layer separation in regions where the actual boundary layer is reasonably well-behaved. In particular, the unsatisfactory behavior of the analysis, when applied to a flat plate followed by a curved compression surface, has been demonstrated by McLafferty and Barber.⁶ Reference 6 has also presented an empirical method of analysis which provides reasonable agreement with their experimental results.

The analysis presented herein attempts to eliminate some shortcomings of the earlier methods by incorporating more recently available results of turbulent boundary-layer analyses. This analysis is similar, in some respects, to the work of Ref. 5; the boundary-layer momentum integral equation is used in conjunction with the moment-of-momentum equation. The wall shearing stress is given by the Ludwig-Tillmann⁷ formulation. However, unlike previous investigations no attempt is made to uncouple the momentum and moment-of-momentum integral equations. Instead, the set of two differential equations is solved simultaneously after introduction of normalized boundary-layer shearing stress distributions. The shear distributions are established from numerical results of equilibrium turbulent boundary-layer analysis.

II. Development of Analysis

A. Basic Equations

For steady, compressible turbulent flow the Prandtl boundary-layer equations, in which flow variables appear as time-averaged quantities, are expressible by

$$\partial(\rho u)/\partial x + \partial(\rho v)/\partial y + j(\rho u/R)(dR/dx) = 0 \quad (1)$$

and

$$\rho u(\partial u/\partial x) + \rho v(\partial u/\partial y) = -(dp/dx) + \partial\tau/\partial y \quad (2)$$

To simplify the final boundary-layer integral equations a Mager-type⁸ transformation is employed, which is of the form

$$X = \int_0^x b(x) dx \quad (3)$$

and

$$Y = c(x) \int_0^y \frac{\rho}{\rho_0} dy \quad (4)$$

where

$$b = (T_0/\bar{T})(T_e/T_0)^{(\gamma+1)/2(\gamma-1)} \quad c = (T_e/T_0)^{1/2}$$

and where X and Y represent the transformed coordinates. Consistent with the transformation in which $\Psi \doteq \psi$, it is useful to observe that

$$u = cU \quad H = [1 + (\gamma - 1)M_e^2/2]H_v + (\gamma - 1)M_e^2/2 \\ \theta = \Theta(T_0/T_e)^{(\gamma+1)/2(\gamma-1)}$$

and

$$\delta^* = (\Theta + \Delta^*)(T_0/T_e)^{(3\gamma-1)/2(\gamma-1)} - \Theta(T_0/T_e)^{(\gamma+1)/2(\gamma-1)}$$

It may be recalled that the functional dependence of $b(x)$ and $c(x)$ for the transformation of form (3) and (4) remains arbitrary until several additional conditions are imposed beyond the initial requirement that the Jacobian of the transformation be nonvanishing, and likewise that requirements are generally dictated such that the resulting equations are simplified if not made completely analogous to the incompressible flow equations. In this context it may be observed that the foregoing explicit transformation differs slightly from that used by Mager in that he postulates that the apparent turbulent shear stress associated with an elemental mass remain invariant under transformation; this postulate has been subject to question, as commented on, for example, by Coles.⁹

The transformed momentum integral equation (5) follows from the governing equations (1) and (2) after applying the preceding transformation.

$$\frac{d\Theta}{dX} + \frac{\Theta}{U_e} \frac{dU_e}{dX} \left[2 + H_i + \frac{1}{\Theta} \int_0^\Delta (g - 1) dY \right] + \\ j \frac{\Theta}{R} \frac{dR}{dX} = \left(\frac{T_0}{T_e} \right) \left(\frac{\bar{T}}{T_0} \right) \frac{\tau_w}{\rho_e U_e^2} \quad (5)$$

where

$$\Theta = \int_0^\Delta \frac{U}{U_e} \left(1 - \frac{U}{U_e} \right) dY$$

and

$$H_i = \frac{1}{\Theta} \int_0^\Delta \left(1 - \frac{U}{U_e} \right) dY$$

In a manner similar to Ref. 5, the accompanying moment-of-momentum integral equation corresponding to Eq. (5) is obtained. Although the general derivation procedure is given in Ref. 5 for completeness, the mechanics of the development shall be restated here briefly; the differential form of Eq. (5) is multiplied by Y , and the result integrated with respect to Y from $Y = 0$ to $Y = \Delta$. To aid in the reduction of the resulting equation to manageable form, the results of Tetervin and Lin³ are utilized. It may be observed that the resulting Eq. (6) is ostensibly the same as that of Ref. 5, except for the modification of the transformation employed:

$$\frac{dH_i}{dX} = - \frac{1}{U_e} \frac{dU_e}{dX} \left[\frac{H_i(H_i + 1)^2(H_i - 1)}{2} \right] \times \\ \left[1 + \frac{2}{(H_i + 1)\Theta} \int_0^\Delta (g - 1) dY - \right. \\ \left. \frac{2(H_i - 1)}{H_i^2(H_i + 1)\Theta^2} \int_0^\Delta (g - 1) Y dY \right] + \frac{H_i(H_i^2 - 1)}{\Theta} \times \\ \left(\frac{T_0}{T_e} \right) \left(\frac{\bar{T}}{T_0} \right) \left(\frac{\tau_w}{\rho_e U_e^2} \right) - \frac{(H_i^2 - 1)(H_i + 1)}{\Theta} \times \\ \left(\frac{T_0}{T_e} \right) \left(\frac{\bar{T}}{T_0} \right) \frac{\tau_w}{\rho_e U_e^2} \int_0^1 \frac{\tau}{\tau_w} d \left(\frac{Y}{\Delta} \right) \quad (6)$$

B. Boundary-Layer Shear Relationships

The Ludwig-Tillman skin-friction relation for incompressible turbulent flow was selected for use in the analysis. For, in addition to its convenient formulation, the expression for

C_f is valid over a wide range of pressure gradients as various authors have noted.¹ The formulation for incompressible flow from Ref. 7 is given by

$$C_f = \tau_w / \frac{1}{2} \rho u_e^2 = 0.246e^{-1.561H_i} Re_\theta^{-0.268} \quad (7)$$

The corresponding skin-friction coefficient for compressible flow is obtained upon application of the reference enthalpy concept. As noted by Tetervin,¹⁰ the reference enthalpy method is quite easily applied to a C_f law with a Re_θ dependence without resorting to a flat-plate momentum equation when it is recognized that θ likewise contains a base density. Indeed, if a flat-plate equation were used, the reference method might appear restricted to zero pressure gradient flows.

Therefore, adapting Eq. (7) for compressible flow

$$\tau_w / \bar{\rho} u_e^2 = 0.123e^{-1.561H_i} (u_e \bar{\theta} / \bar{\nu})^{-0.268}$$

And since $\bar{\rho} / \rho_e = T_e / \bar{T}$

$$\tau_w / \rho_e u_e^2 = 0.123e^{-1.561H_i} (u_e \bar{\theta} / \bar{\nu})^{-0.268} (T_e / \bar{T})^{1.268} \quad (8)$$

where $(\bar{})$ refers to conditions evaluated at the reference temperature given by Eckert,¹¹

$$\bar{T} / T_0 = 0.5 T_w / T_0 + 0.22 \sigma^{1/3} + (0.5 - 0.22 \sigma^{1/3}) (T_e / T_0) \quad (9)$$

Since $\theta = (T_0 / T_e)^{3/4} \Theta$ and $u_e = (T_e / T_0)^{1/2} U_e$ for $\gamma = 1.4$, then

$$C_f / 2 = \tau_w / \rho_e u_e^2 = 0.123e^{-1.561H_i} (U_e \Theta / \nu_0)^{-0.268} \times (T_e / \bar{T}) (\bar{\mu} / \mu_0)^{0.268} \quad (10)$$

Evaluation of the normalized boundary-layer shear distribution arising in Eq. (6) is also required for the solution of Eqs. (3) and (4). To this end, the results of equilibrium turbulent boundary-layer analysis are incorporated. In particular, as shown by Libby et al.,¹²

$$H_i = \left\{ 1 + \left[\left(\frac{C_f'}{2} \right)^{1/2} \int_0^1 F^2 d\eta / \int_0^1 F d\eta \right] \right\}^{-1} \quad (11)$$

where $F = u_e / u_\tau [(u / u_e) - 1]$ and $\eta = y / \delta$ for an incompressible flowfield. If a wide range of both adverse and favorable equilibrium pressure distributions are considered, then the shear integral may be expressed in the approximate form

$$\int_0^1 \frac{\tau}{\tau_w} d\eta = 0.011 \left[\frac{H_i - 1}{H_i} \right]^2 \left(\frac{C_f'}{2} \right)^{-1} \quad (12)$$

where C_f' is the appropriate skin-friction coefficient for the incompressible flow. In this sense, care should be exercised in relating C_f' to the flowfield it is assumed to represent.

The functional dependence of

$$\int_0^1 \frac{\tau}{\tau_w} d\eta$$

upon H_i as given by the preceding analysis is presented in Fig. 1. Also shown is the shear distribution assumed in Ref. 5; this has a much smaller range of validity due to, among other reasons, the assumed functional dependence only upon H_i . The available experimental results (cf. Refs. 13-15) are also included in Fig. 1. All the data have been obtained for an incompressible flow, and in the majority of cases adverse pressure gradients are present. As may be noted, there exists an underabundance of detailed experimental turbulent boundary-layer data even for the incompressible case and essentially none for the compressible case.

C. Extension to Nonadiabatic Flow

For unity Prandtl number, the integrals

$$\int_0^\Delta (g - 1) dY \quad \text{and} \quad \int_0^\Delta (g - 1) Y dY$$

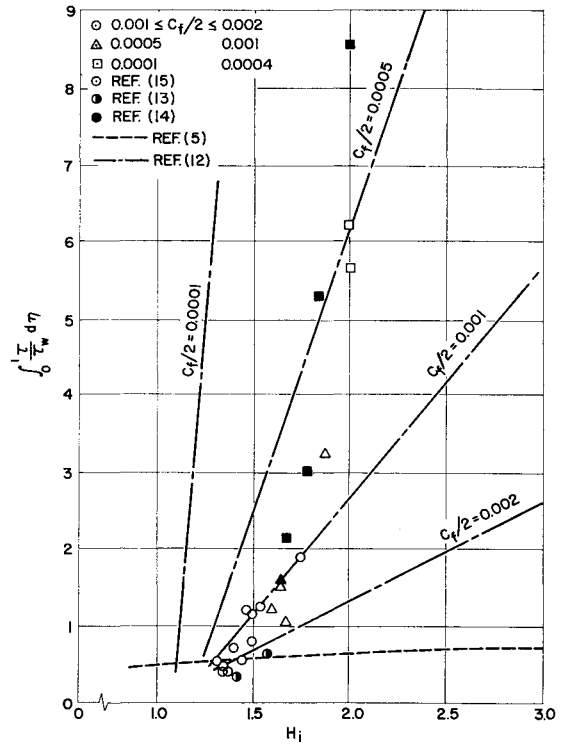


Fig. 1 Shear integral for incompressible turbulent boundary layers.

arising in Eqs. (5) and (6) are equal to zero for adiabatic flow. To evaluate these integrals for an arbitrary but constant surface temperature, the Crocco relation is employed with a power law assumption for the velocity profiles, assuming the Prandtl and Lewis numbers are unity. Specifically, when

$$(h_s - h_w) / (h_0 - h_w) = u / u_e = U / U_e$$

$$U / U_e = (Y / \Delta)^{N_i}$$

then

$$\int_0^\Delta (g - 1) dY = (g_w - 1) H_i \Theta \quad (13)$$

and

$$\int_0^\Delta (g - 1) Y dY = (g_w - 1) \frac{H_i^2 (H_i + 1)^2}{2(H_i - 1)(H_i + 3)} \Theta^2 \quad (14)$$

where

$$H_i = 2N_i + 1$$

It is interesting to note that a Mager-type transformation causes increased distortion of the velocity profiles with increasing external Mach number. As noted in Ref. 16, if N_i is assumed constant as in a typical power law velocity profile, then the physical profile parameter N will vary with the normal coordinate y .

Although there is some question concerning the applicability of the Crocco relation for flows with heat transfer and pressure gradient, this was utilized in the present analysis in place of solving the energy integral equation directly. A more involved approach would be to solve the energy integral equation simultaneously with the momentum and moment-of-momentum integral equations. This has been done in previous analyses (e.g., Ref. 22), and the results, in general, indicated that the additional complexity did not improve the solution over that obtained by utilizing the Crocco relation. As a result, it is felt that the enthalpy integral evaluation of Eqs. (13) and (14) should be sufficiently accurate for the range of validity of the analysis.

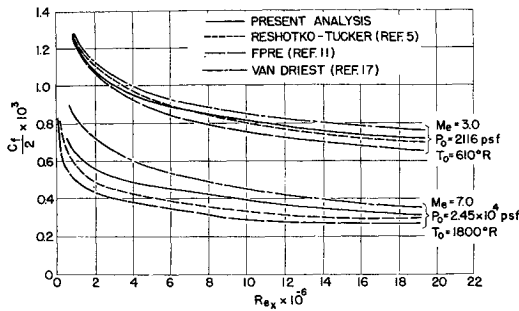


Fig. 2 Skin-friction variation on an adiabatic flat plate ($g_w = 1.0$).

The relation between the actual form factor (H) and the incompressible form factor (H_i) is, therefore, given by

$$H = g_w H_i [1 + (\gamma - 1) M_e^2 / 2] + (\gamma - 1) M_e^2 / 2 \quad (15)$$

Incorporating the results of the preceding sections, the momentum integral equation (5) may be written

$$d\theta/dX + (\theta/U_e)(dU_e/dX)[2 + g_w H_i] + j(\theta/R)(dR/dX) = 0.123e^{-1.561H_i}(U_e\theta/\nu_0)^{-0.268}(T_e/T_0)(\bar{\mu}/\mu_0)^{0.268} \quad (16)$$

and the moment-of-momentum equation (6) may be written

$$\frac{dH_i}{dX} = -\frac{1}{2U_e} \frac{dU_e}{dX} [H_i(H_i + 1)^2(H_i - 1)] \times \left[1 + (g_w - 1) \frac{H_i^2 + 4H_i - 1}{(H_i + 1)(H_i + 3)} \right] + \frac{(H_i^2 - 1)}{\theta} \left(\frac{T}{T_e} \right) \times \left[\frac{H_i \tau_w}{\rho_e U_e^2} - \frac{(H_i + 1) \tau_w}{\rho_e U_e^2} \int_0^1 \frac{\tau}{\tau_w} d\left(\frac{Y}{\Delta}\right) \right] \quad (17)$$

In the past, investigators^{4,5} have uncoupled the momentum equation from the preceding equation set using semiempirical approximations. With the present capability of computing machines, and in order to maintain the coupling effect of H_i and θ , it was deemed highly desirable to retain the momentum equation in the form of Eq. (16) and to solve this numerically in conjunction with Eq. (17).

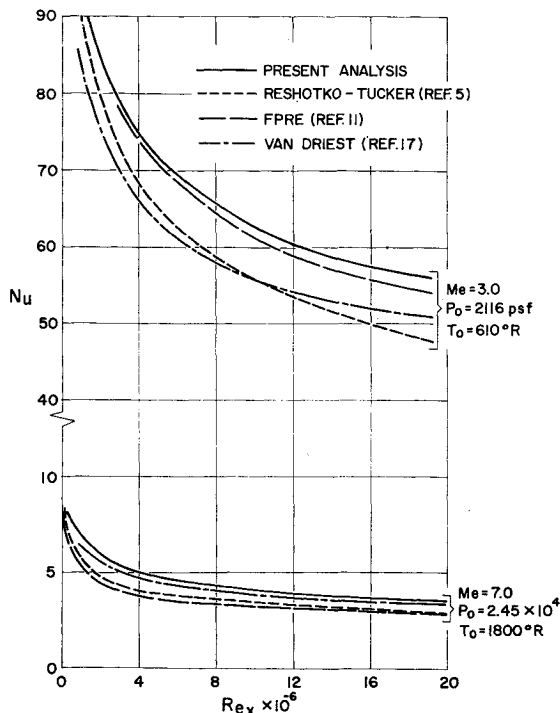


Fig. 3 Nusselt number distribution on a flat plate ($g_w = 0.3$).

It may be seen that both Eqs. (16) and (17) of the equation set appear to have singularities at $\theta = 0$. It can be observed that Eq. (17) has a nonremovable singularity in contrast to Eq. (16), which may be rewritten in the form

$$1.268^{-1}(df/dX) + (f/U_e)(dU_e/dX)[1 + g_w H_i] + j(f/R)(dR/dX) = 0.123e^{-1.561H_i}(T_e/T_0)(U_e/\nu_0)(\bar{\mu}/\mu_0)^{0.268} \quad (18)$$

where

$$f \doteq (U_e\theta/\nu_0)^{1.268}$$

Equations (17) and (18) may also be expressed in the form

$$df/dx = 1.268 \{ (-f/M_e)(dM_e/dx)[(1 + g_w H_i) - (jy/R)(dR/dx) + A] \} \quad (19)$$

and

$$\frac{dH_i}{dx} = -\frac{1}{2M_e} \frac{dM_e}{dx} [H_i(H_i + 1)^2(H_i - 1)] \times \left[1 + (g_w - 1) \frac{H_i^2 + 4H_i - 1}{(H_i + 1)(H_i + 3)} \right] + \frac{(H_i^2 - 1)}{f} \times A \left[H_i - \frac{0.011(H_i + 1)(H_i - 1)^2}{H_i^2} \cdot \frac{2}{C_f} \frac{T_0}{T} \right] \quad (20)$$

where

$$f = (M_e a_0 \theta / \nu_0)^{1.268}$$

and

$$A = 0.123e^{-1.561H_i}(M_e a_0 / \nu_0)(T_e/\bar{T})(T_e/T_0)^3(\mu/\mu_0)^{0.268}$$

Hence, upon specification of a body shape and the accompanying external flowfield, simultaneous solution of the preceding differential equations yield $H(x)$ and $\theta(x)$ for the boundary layer in question. Typical initial conditions for these equations are discussed in the numerical examples that follow.

III. Comparison of Analysis with Theoretical and Experimental Results

Application of the analysis and comparison with other theoretical results is afforded by several examples given below.

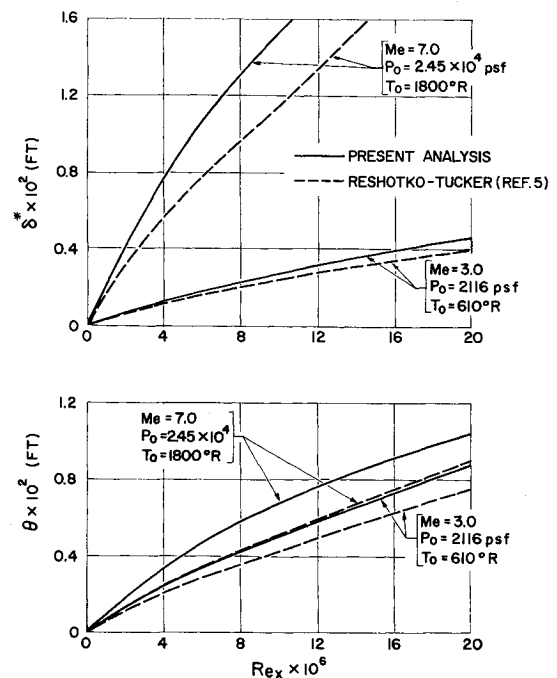


Fig. 4 Variation of integral thicknesses on a flat plate ($g_w = 1.0$).

The numerical results are obtained by a Runge-Kutta integration technique programmed on a Bendix G-15 digital computer. Reference quantities are determined by Eckert's reference enthalpy formulation for Prandtl number equal to 0.72; viscosity ratios are determined by the Sutherland formula.

A. Flow over Flat Plates

Adiabatic and nonadiabatic flat-plate flows for Mach numbers of 3 and 7 are examined in Figs. 2-5. Inasmuch as one of the integral equations is singular at $x = 0$, H_i was set equal to 1.3 in the vicinity of the leading edge. At a small distance downstream of the leading edge both equations were solved simultaneously, yielding rapid convergence to the actual value of H_i .

For $g_w = 1$, Fig. 2 shows the variation of skin friction with Re_x obtained from different analyses. For comparison, the methods of Reshotko-Tucker,⁵ van Driest,¹⁷ and the flat-plate reference enthalpy method¹⁸ (FPRE) are also presented. The van Driest formulation using the Prandtl mixing length hypothesis is employed. All of the analyses shown are in close agreement at low external Mach numbers but tend to diverge as the Mach number increases.

For $g_w = 0.3$, Fig. 3 shows the Nusselt number dependence on Re_x . Here, as in the remainder of the examples, use is made of Reynolds' analogy for $\sigma = 0.72$. Again, it is noted that generally close agreement is obtained for all analyses.

The momentum and displacement thickness for the various flat plates considered are given in Figs. 4 and 5. Dependence of θ and δ^* on Re_x are compared with the predictions of Ref. 5 under the same flow conditions. It is noted that as external Mach numbers increase, the variation in δ^* and θ between the results of Ref. 5 and the present analysis becomes more pronounced, although qualitatively the behavior of the boundary-layer thicknesses is similar.

B. Axisymmetric Flow with Mild Adverse Pressure Gradient

Zakkay and Callahan¹⁹ obtained, among others, experimental heat-transfer data over a cone-cylinder-flare configuration shown schematically in Fig. 6. A mild adverse

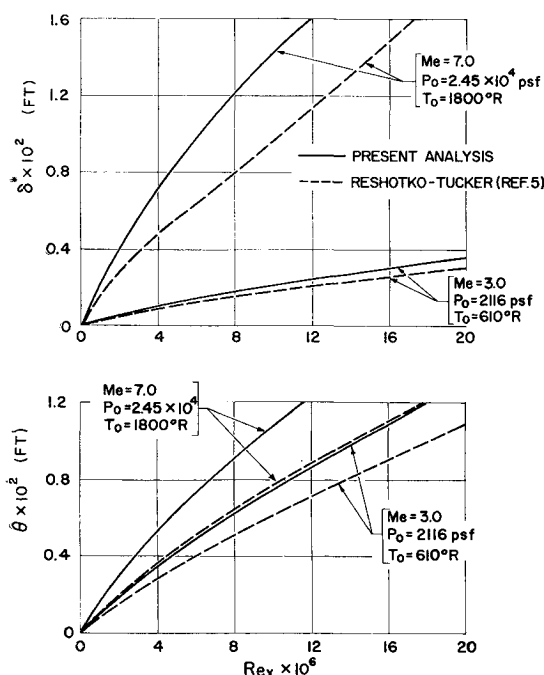


Fig. 5 Variation of integral thicknesses on a flat plate ($g_w = 0.3$).

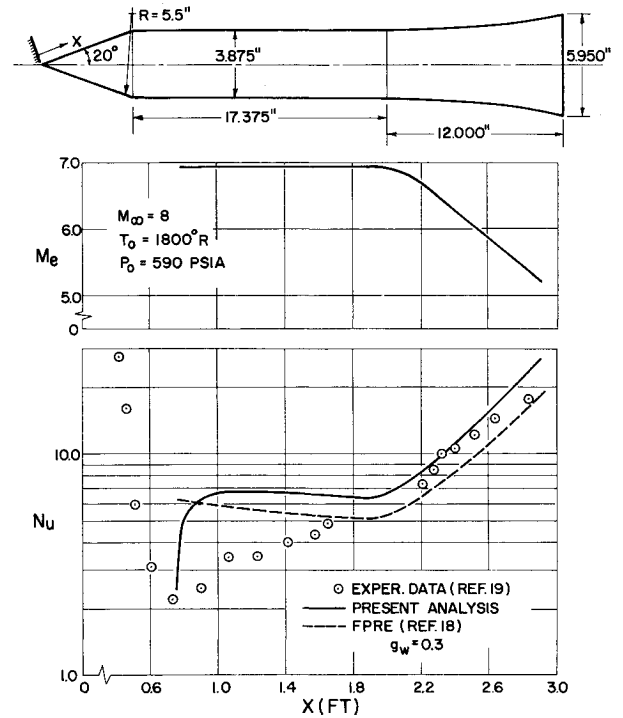


Fig. 6 Mach number and Nusselt number distributions on axisymmetric configuration.¹⁹

pressure gradient is present on the aft portion of the model as indicated by the resulting Mach number distribution given in Fig. 6.

The present analysis is applied to the configuration of Ref. 19 using the test condition corresponding to the largest extent of fully turbulent flow achieved in the test series. These conditions are presented in Fig. 6. The analysis is initiated in the laminar-transitional region of the body, using an initial value of momentum thickness computed from laminar theory in Ref. 19. Although the starting value of H_i equal to 1.9 is that associated with a laminar boundary layer, the solution rapidly converges to the fully developed turbulent boundary-layer solution; this is readily seen in the plot of Nusselt number vs surface distance (Fig. 6). The heat transfer predicted by the present analysis rapidly converges to values expected from a fully turbulent boundary-layer flow, as indicated by the FPRE method. It is seen that on the aft portion of the body, where the flow is fully turbulent, good agreement is obtained between experimental data and the present analysis. It is also significant that the behavior of the solution indicates that the results are relatively insensitive to the starting values used in the analysis.

Here, Reynolds analogy has again been employed in the form

$$Nu = \sigma^{1/3} [(h_{aw} - h_w)/(h_0 - h_w)] (\rho_e u_e L / \mu_0) (C_f / 2) \quad (21)$$

where $\sigma^{1/3}$ is the empirical correction for nonunity Prandtl number, and where the adiabatic wall enthalpy (h_{aw}) is defined in terms of the recovery factor

$$(h_{aw} - h_e)/(h_0 - h_e) = \sigma^{1/3}$$

Equation (21) has been applied in this case to the pressure-gradient flow of Ref. 19 to provide a means of comparison. It was found, as in Ref. 22, that the accuracy of Nu prediction did not increase significantly for calculations made by extending Reynolds analogy to include pressure gradient terms.

C. Flows with Strong Adverse Pressure Gradient

The adiabatic flows of Ref. 6 are of particular interest for they contain large adverse pressure gradients, as can be seen

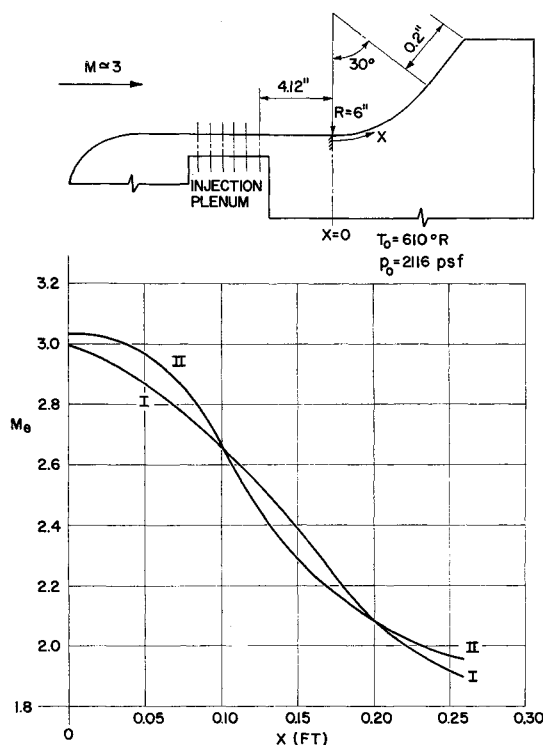


Fig. 7 Two-dimensional model configuration and Mach number distribution.⁶

from Fig. 7. The two flowfields shown, denoted in the present figures by flow I and II, respectively, were obtained by compressing an initially uniform Mach 3 flow over a flat plate by insertion of a compression surface. The geometry of the two-dimensional tunnel model is also shown in Fig. 7. To vary the boundary-layer thickness at $x = 0$, a small amount of air was injected into the boundary layer upstream of the surface under consideration. Additional details can be found in Ref. 6.

Employing the experimentally determined external flowfields and initial conditions cited by this reference, the result-

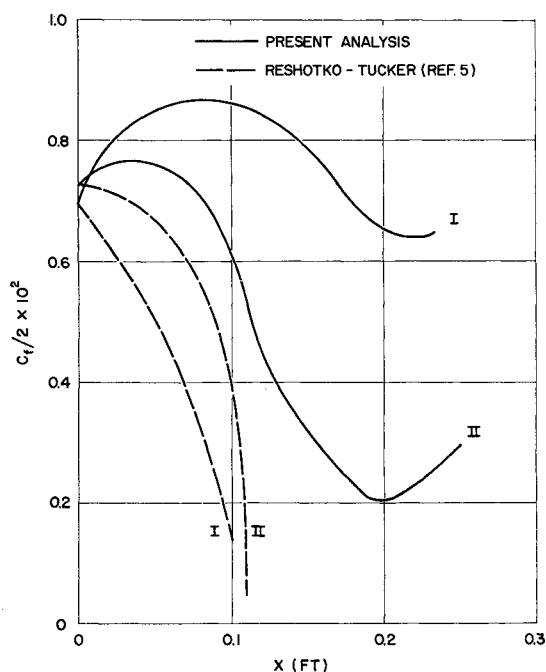


Fig. 8 Variation of skin-friction coefficient on two-dimensional model.

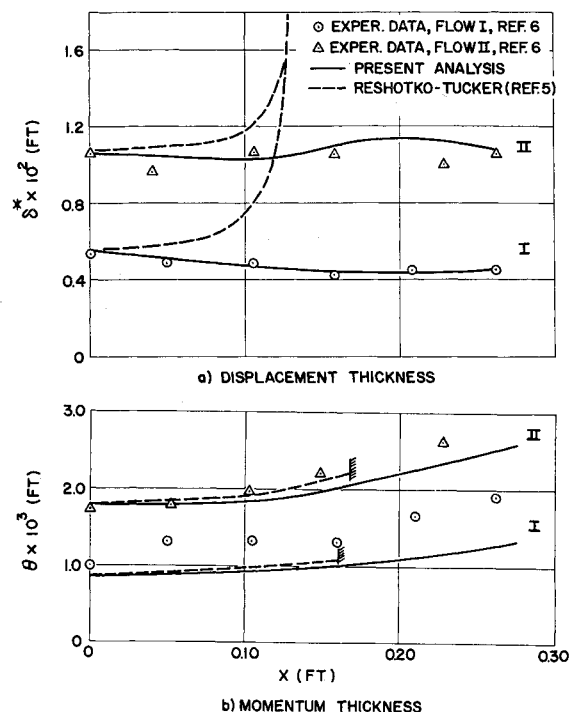


Fig. 9 Variation of boundary-layer thicknesses on two-dimensional model.

ing variations of H_i and Θ and consequently θ , δ^* , and $C_f/2$ are obtained by the present analysis. The skin friction and boundary-layer thicknesses are presented in Figs. 8 and 9, along with the experimental data for flow I and II. Examination of Figs. 9a and 9b reveals good agreement between the analysis and the experimentally measured values of δ^* and θ . Again, for purposes of comparison with other methods, results of the Reshotko-Tucker analysis are also included. Although no direct measurement of C_f was made in Ref. 6, by inspection of the $C_f/2$ vs x curves in Fig. 8, it is seen that the present analysis yields a variation in skin-friction coefficient

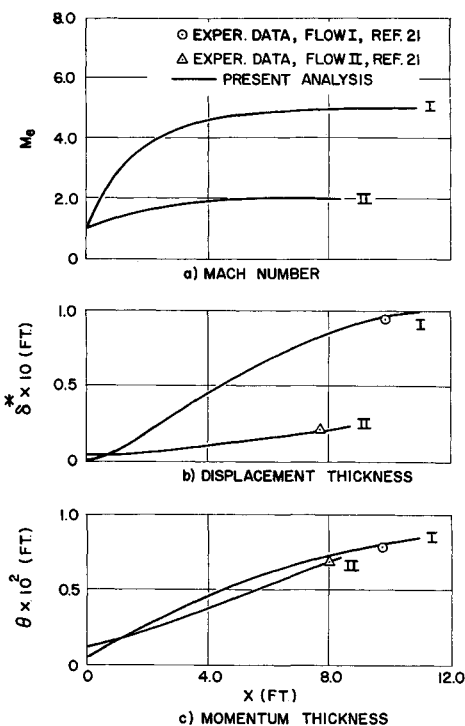


Fig. 10 Variation of Mach number and boundary-layer thicknesses on a nozzle.²⁰

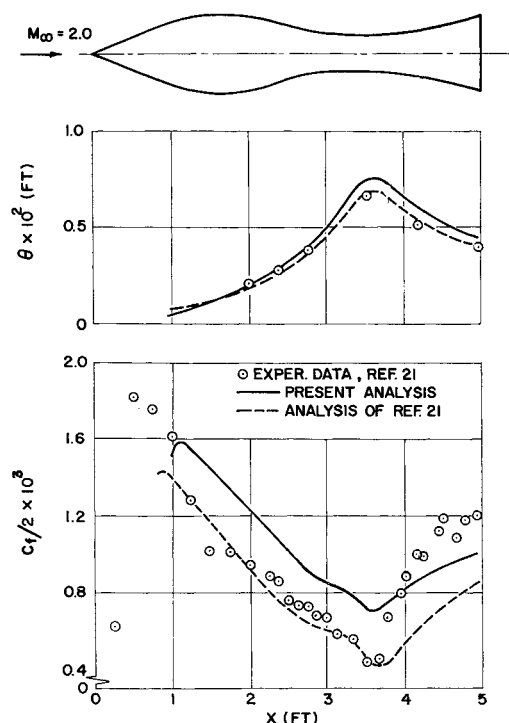


Fig. 11 Variation of skin friction and momentum thickness on axisymmetric configuration.²¹

that does not approach zero and, therefore, is consistent with the experimental observation that separation did not occur for this flow.

D. Flows with Favorable Pressure Gradient

Several favorable and favorable-adverse pressure gradient flows were also considered. The first case in point is represented by data from two California Institute of Technology Jet Propulsion Laboratory supersonic nozzles, as shown in Fig. 10. Calculations were initiated at the nozzle throats using initial θ values listed in Ref. 20. It may be seen that both δ^* and θ agree closely with the experimental points obtained by Dayman²⁰ for the Mach 5 and the Mach 2 adiabatic nozzles.

Secondly, Mach 2 flow past an axisymmetric waisted body of Ref. 21 is examined (Fig. 11). Calculation was initiated in the boundary-layer transition region. Again, it may be observed that θ is in good agreement with data points, as is $C_f/2$, which falls within the range of experimental error stated in Ref. 21. Also shown are the analysis curves of this reference. The analysis curves were obtained by approximating the very difficult task of simultaneous integration of both momentum and energy integral equations through preliminary correlations of the adiabatic data to be analyzed subsequently. Such an analysis, in which direct consideration is given to the energy equation, might indeed be promising if the initial difficulties of this approach can be transcended. As shown in Ref. 21, extension of such an approach to other than adiabatic flows within a small Mach number range appears to be reasonably difficult.

IV. Conclusions

The results of the present analysis indicate that the simultaneous solution of the boundary-layer integral equations, without attempting to uncouple the same, may indeed be a possible method for treatment of the general compressible turbulent boundary layer. The present method yields skin-friction coefficients and other boundary-layer parameters in close agreement with experimental and theoretical results over a wide range of zero, favorable, and adverse pressure-gradient

flows. Favorable results for nonadiabatic flows with zero and mild pressure gradients are also obtained.

In addition, it was found that this method of analysis did not give spurious indication of separation in the presence of large adverse pressure gradients, but yielded boundary-layer parameters in good agreement with experimentally obtained data.

References

- 1 Clauser, F., "The turbulent boundary layer," *Advances in Applied Mechanics* (Academic Press Inc., New York, 1956), Vol. IV.
- 2 Von Doenhoff, A. E. and Tetervin, N., "Determination of general relations for the behavior of turbulent boundary layers," NACA Rept. 772 (1943).
- 3 Tetervin, N. and Lin, C. C., "A general integral form of the boundary-layer equation for incompressible flow with an application to the calculation of the separation point of turbulent boundary layers," NACA Rept. 1046 (1951).
- 4 Maskell, E. C., "Approximate calculation of the turbulent boundary layer in two-dimensional incompressible flow," British Royal Aircraft Establishment Rept. Aero 2443 (1951).
- 5 Reshotko, E. and Tucker, M., "Approximate calculation of the compressible turbulent boundary layer with heat transfer and arbitrary pressure gradient," NACA TN 4154 (1957).
- 6 McLafferty, G. H. and Barber, R. E., "The effect of adverse pressure gradients on the characteristics of turbulent boundary layers in supersonic streams," *J. Aerospace Sci.* **29**, 1-18 (1962).
- 7 Ludwig, H. and Tillmann, W., "Investigations of the wall shearing-stress in turbulent boundary layers," NACA TN 1285 (1950).
- 8 Mager, A., Transformation of the compressible turbulent boundary layer," *J. Aerospace Sci.* **25**, 305-311 (1958).
- 9 Coles, D. E., The turbulent boundary layer in a compressible fluid," *Phys. Fluids* **7**, 1403-1423 (September 1964).
- 10 Tetervin, N., "The application of the reference-enthalpy method to friction formulas," *J. Aerospace Sci.* **29**, 493 (1962).
- 11 Eckert, E. R. G., "Engineering relations for friction and heat transfer to surfaces in high velocity flow," *J. Aeronaut. Sci.* **22**, 585-586 (1955).
- 12 Libby, P. A., Baronti, P. O., and Napolitano, P., "Study of the incompressible turbulent boundary layer with pressure gradient," *AIAA J.* **2**, 445-452 (1964).
- 13 Sandborn, V. A. and Slogar, R. J., "Study of the momentum distribution of turbulent boundary layers in adverse pressure gradients," NACA TN 3264 (1955).
- 14 Newman, M. A., "Some contributions to the study of the turbulent boundary-layer near separation," Australian Aeronautical Research Consultative Committee, Rept. ACA-53 (1951).
- 15 Schubauer, G. B. and Klebanoff, P. S., "Investigation of separation of the turbulent boundary layer," NACA TN 2133 (1950).
- 16 Reshotko, E. and Tucker, M., "Effect of a discontinuity on turbulent boundary-layer thickness parameters with application to shock-induced separation," NACA TN 3454 (1955).
- 17 van Driest, E. R., "The problem of aerodynamic heating," *Aeronaut. Eng. Rev.* **15**, 26-41 (October 1956).
- 18 Cresci, R. J., MacKenzie, D. A., and Libby, P. A., "An investigation of laminar, transitional, and turbulent heat transfer on blunt-nosed bodies in hypersonic flow," *J. Aerospace Sci.* **27**, 401-414 (1960).
- 19 Zakkay, V. and Callahan, C. J., "Laminar, transitional and turbulent heat transfer to a cone-cylinder-flare body at Mach 8.0," *J. Aerospace Sci.* **29**, 1403-1420 (1962).
- 20 Dayman, B., "Comparison of calculated with measured boundary-layer thicknesses on the curved walls of the JPL 20 inch supersonic wind tunnel two-dimensional nozzle," California Institute of Technology Jet Propulsion Laboratory TR 32-349 (March 1963).
- 21 Winter, K. G., Smith, K. G., and Rotta J. C., "Turbulent boundary-layer studies on a waisted body of revolution in subsonic and supersonic flow," AGARDograph 97, Pt. II (May 1965).
- 22 Libby, P. A. and Cresci, R. J., "Evaluation of several hypersonic turbulent heat transfer analyses by comparison with experimental data," Polytechnic Institute of Brooklyn Aerospace Lab. PIBAL 387, Wright Air Development Center WADC TN 57-72 (July 1957).